



Voting System Anomaly Root Cause Analysis Template v2.0

Root Cause Analysis for:

**VV40ECT-74: VxSCAN RADIATED ELECTROMAGNETIC EMISSIONS (CLASS B) FAILURE AT 54 MHz AND
66 MHz - REV. 3
VxSUITE, VERSION 4.0 AND EAC CERTIFICATION #VXS4**

**VOTINGWORKS
548 MARKET ST, STE 53001
SAN FRANCISCO, CA 94104-5401**

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Introduction

This document describes the cause of failure of Radiated Electromagnetic Emissions testing (IEC61000-4-3 (2020) Level 3) from 30 MHz to 1 GHz in the certification process with SLI Compliance and Element in March 2025, for VxScan SC-11 hardware. In that test, excess RF emissions were detected in the lower range of the spectrum, and even when using an alternate backup uninterruptible power supply (UPS) and other mitigations, problematic spikes were detected at approximately 54 and 66 MHz. Previous RF emissions testing conducted on VxScan in July 2024 with NOVO Engineering and Nemko did not show this issue. An investigation of the issue was then important to narrow down its cause at SLI/Element and prevent it in the design going forward. Updated SC-15 VxScan hardware addresses these issues.

Anomaly Description

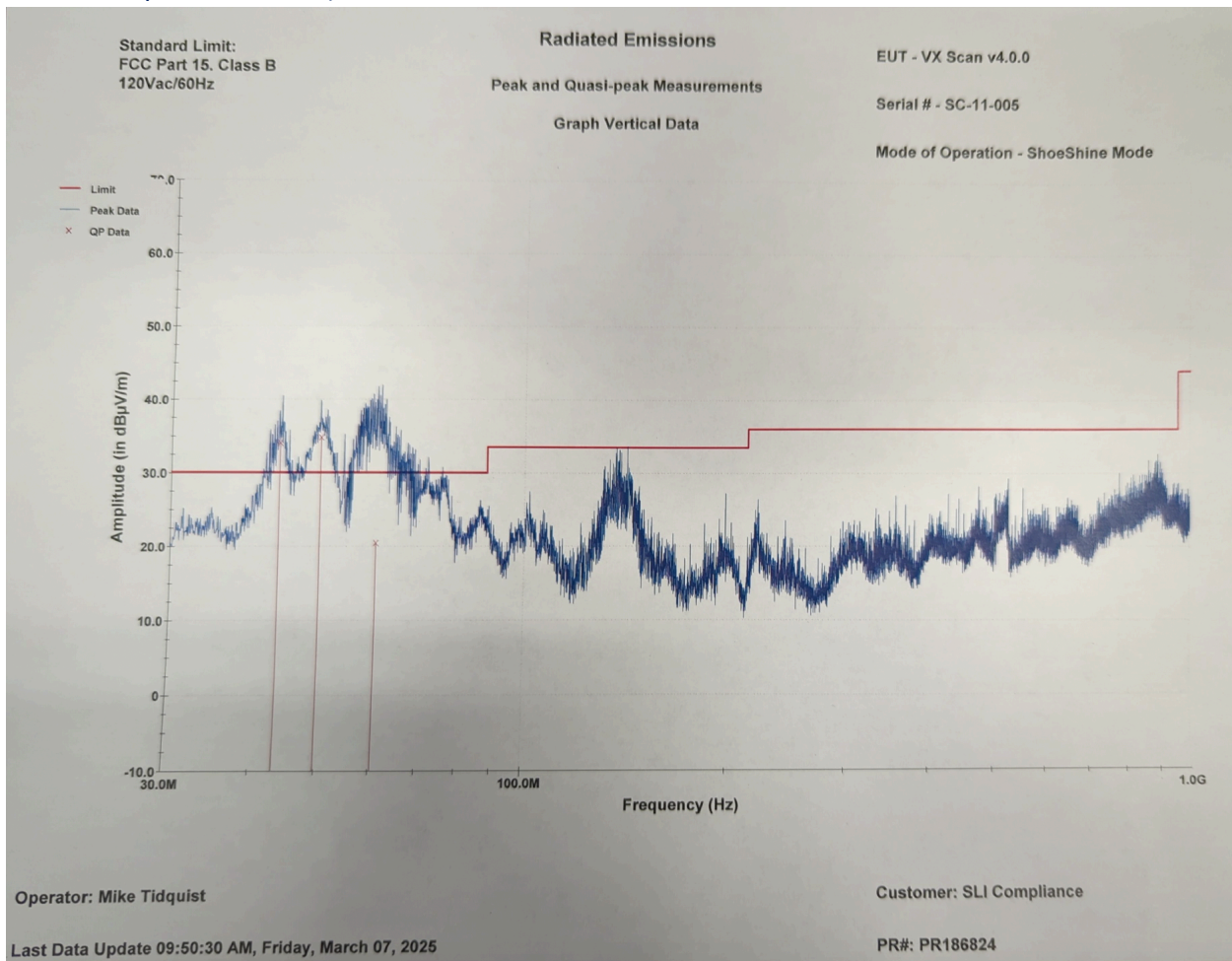
Complete all sections. Descriptions must be as detailed as possible, while being clear and concise since the anomaly is the source of the entire RCA. This detail should include a complete list and/or description of the “symptoms” of the anomaly and the conditions present which the symptoms occurred.

<u>Date of Anomaly:</u> March 7, 2025	<u>Time of Anomaly:</u> 9 AM (Mountain Time)
<u>Place of Anomaly:</u> Element, Longmont, CO	<u>Person identifying Anomaly:</u> Jessica Myers, VotingWorks
<u>Expected Results of actions leading up to anomaly:</u> VxScan and its original UPS (Anker Solix C300) were set up to undergo standard tests for radiated emissions (FCC Part 15, Class B) at Element in Longmont, Colorado with SLI Compliance. The system was expected to pass emissions tests without issue, based on past EMC testing with NOVO Engineering and Nemko in July 2024 where VxScan did not show concerning emissions. The tests at Element had slightly different conditions from tests in July 2024 without the Anker UPS, but emissions were not expected to be worse with the UPS.	

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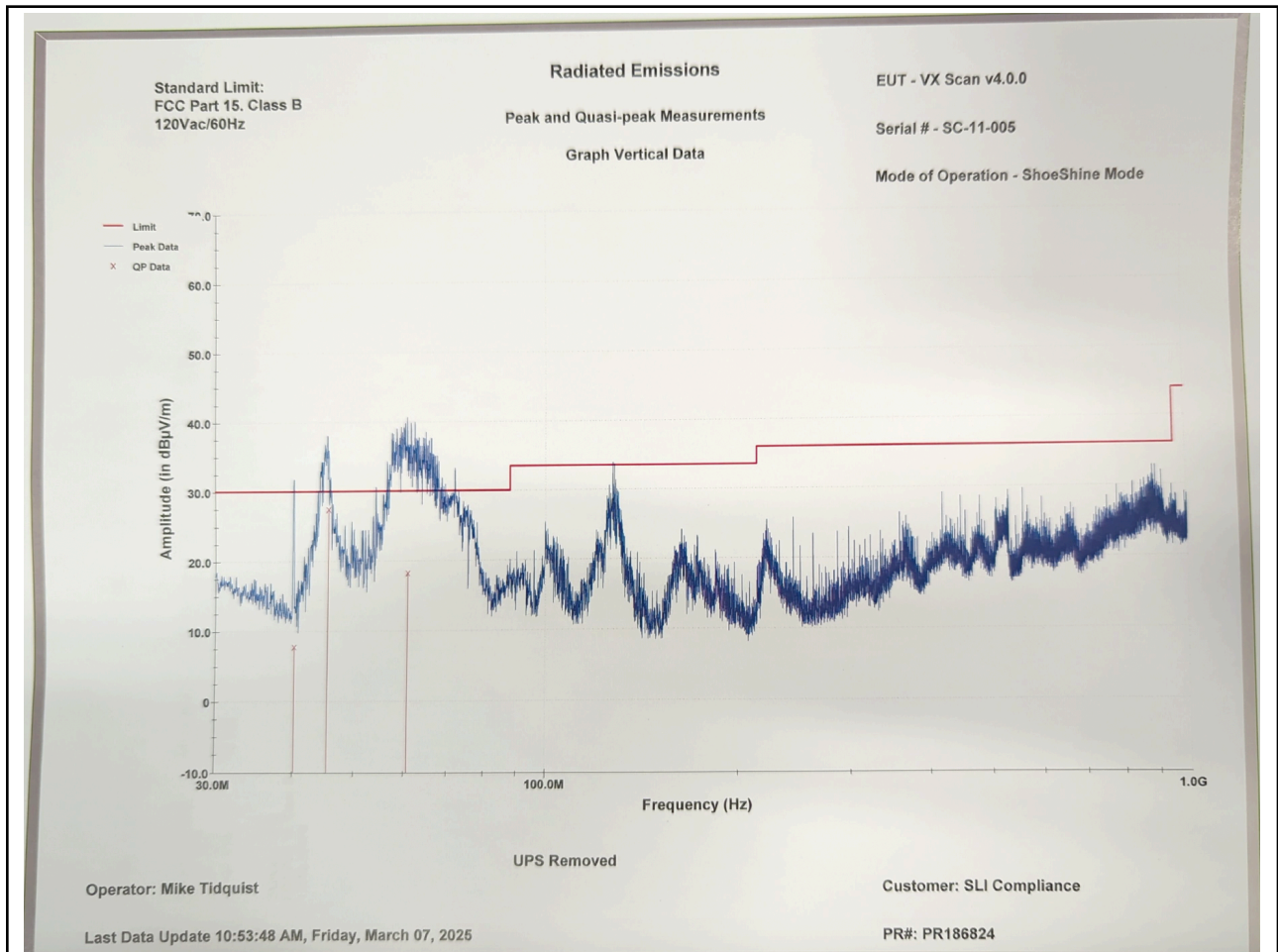
Detailed description of the event / anomaly:

Instead of passing emissions tests, the VxScan system when connected with the Anker UPS failed Radiated Emissions testing from 30 Mhz to 1 GHz. Since this was unexpected, a series of additional tests followed to confirm emissions behavior of the system, showing that the UPS appeared to show problematic emissions. This can be seen in emissions data from when VxScan was powered through the UPS connected to mains (first graph below, QP amplitudes exceeding desired amplitudes in red):



Powering VxScan directly through mains and no UPS also showed some noisy emissions in a similar frequency range, but the QP amplitudes did not exceed emissions limits in this case. So connecting the Anker UPS appeared to contribute to the excess emissions (2nd graph below, exceeding desired amplitudes in red):

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Applying mitigations, such as replacing the Anker UPS with an alternate APC-brand UPS (APC Back-UPS Pro, BN1500M2), still appeared to show excess emissions in the lower frequency band, with spikes around 54 and 66 MHz, although QP data were not available.

If the anomaly is repeatable, provide step by step instructions to recreate it:

1. Image VxScan with a test software ("Hardware Test Utility") that continuously runs its major components, including running the scanner in shoeshine mode.
2. Set up the VxScan system to be powered from the Anker UPS, which in turn is powered by plugging directly into mains.
3. Power on VxScan and run the test software utility. Insert a sheet of paper, thermal paper, USB stick, and smart card into VxScan to implement shoeshine mode.
4. Measure radiated emissions from the system according to FCC Part 15, Class B, standards.

Chronology of Events / Timeline

Provide a detailed chronology of the events leading up to, and following, the anomaly. Add additional events if necessary.

ID	Date/Time	Description	Entity Org/person	Result / Notes
1	3/7/25, 8:30am (Mountain Time)	VxScan + Anker UPS was scanned for radiated emissions, 30 MHz - 1 GHz, according to FCC Part 15, Class B, standards, in standard certification testing. Two ferrites were clipped onto the scanner USB data cable to the SBC, to address previous successful radiated RF immunity mitigations that led to passed tests.	Jessica Myers, VotingWorks, with SLI Compliance and Element	System failed emissions test, with excess emissions around 40-65 MHz, especially in the vertical direction. Initial investigations with a near-field probe indicated ~50 MHz noise from the Anker UPS near the main control panel and screen. The VxScan system also showed a source of noise around the right side of the scanner access panel door, near the printer stepper motor. It was also noted that this noise was present with the UPS wireless functionality switched off according to the user-controlled IoT button on the front. If the IoT button were on, more emissions would be expected in the higher frequency range. Plans were made to retest using an alternate UPS (APC) already used for VxMarkScan, another product in the VxSuite family.
2	3/10/25, 12:20pm	VxScan + APC UPS was scanned for radiated emissions, 30 MHz - 1 GHz.	Jessica Myers, Don Chu, VotingWorks, with SLI Compliance and Element	System failed emissions test, with excess QP emissions spikes at about 54 MHz and 66 MHz. Overall noise across the spectrum was lower than in the Anker tests, and the APC data was noticeably cleaner. A near-field probe was used to analyze more sources of emissions around the system. Plans were made to test additional mitigations to the VxScan unit itself to the printer motor housing and cables and the touchscreen control electronics, in an attempt to attenuate potential sources of the spikes.
3	3/10/25, 2:38pm	VxScan + APC UPS, with more mitigations, was scanned for radiated emissions, 30 MHz - 1 GHz.	Jessica Myers, Don Chu, VotingWorks, with SLI Compliance and Element	Data was collected on these potential mitigations with the APC UPS, although no QP data was collected to definitively say if the mitigations could have passed. Similar spikes were seen in the lower frequency range.

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				Plans were made to (1) continue other EMC tests at Element to analyze the system and the Anker UPS, and (2) do deeper investigations on VxScan emissions with an outside party: NOVO Engineering in San Diego. The initial focus of NOVO was to analyze the UPS and the printer, since data from SLI/Element suggested those were sources of problematic emissions.
4	3/31/25	NOVO Engineering finished conducting EMC tests of interest on another VxScan unit, along with a new Anker UPS, the APC UPS model, and a third potential UPS from Goldenmate, sharing findings with VotingWorks.	Al Walters, Jeff Johnson, NOVO Engineering, with Pius Wong VotingWorks	Tests confirmed the Anker UPS should change to another model. The Anker UPS failed Voltage Drop tests, while the alternate UPS models (APC and Goldenmate) passed. Minimal differences in the UPS's were found in the other tests of emissions, EFT, and voltage interruption. Emissions were analyzed with a near-field probe, with plans to investigate it further according to FCC Class B standards with test vendor Nemko.
5	4/1/25	The VxScan units at Element/SLI were examined to assess the possibility that some cabling features were acting as an antenna and emitting low frequency noise, a known potential failure mode NOVO Engineering had discovered previously in July 2024.	Chris Pedersen, VotingWorks	Three VxScan units at Element/SLI were examined visually and physically (004, 005, and 007). Grounding wires connected to the scanner were confirmed to be connected and routed properly. This suggested that the failure mode NOVO discussed in July 2024 may not be applicable to the current units at Element/SLI, and further investigations of other sources of the noise were needed.
6	4/25/25, 2pm (Pacific Time)	After NOVO Engineering continued EMC testing of VxScan over several weeks, e-field and h-field probe analysis was conducted on different components of the system, while applying potential mitigations especially to printer components.	Mike Engelmann, Al Walters, NOVO Engineering, with Jesse DeWald, VotingWorks	Results did not corroborate or clarify earlier findings about emissions potentially coming from the printer controller board. Data was collected on emissions from the VxScan unit using 2 versions of the Hardware Test Utility: the original version (Hardware Test Utility v1) that used too much paper for each line of text, and the updated version (Hardware Test Utility v2) that unspooled a more realistic and reasonable amount of paper with each printed line. Plans were made to scan VxScan and UPS configurations in 3m and 10m chambers for deeper analysis, to isolate far-field sources of emissions instead of just near-field.

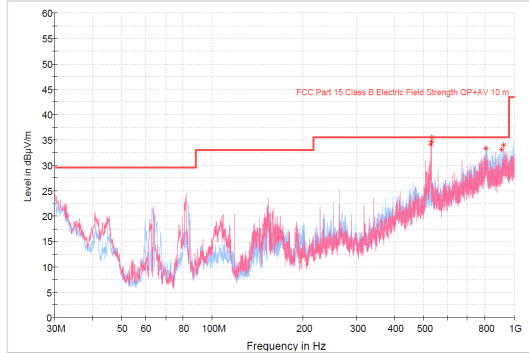
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7	4/28/25, 4pm (Pacific Time)	Standard FCC pre-cert emissions testing was performed on VxScan in different configurations for benchmark data: 1. 3 m chamber, Anker UPS, Hardware Test Utility v1 2. 10 m chamber, Anker UPS, Hardware Test Utility v1 3. 10 m chamber, APC UPS, Hardware Test Utility v1 4. 10 m chamber, APC UPS, Hardware Test Utility v2	Al Walters, Mike Engelmann, NOVO Engineering, and Nemko San Diego	QP data showed consistent excess emissions at 66 MHz, with potential additional emissions of concern nearing FCC limits at other frequencies <80 MHz. Even the 3m chamber test failed at 66 MHz, when the 3m test did not fail previously in July 2024 without a UPS and with different test software. This further suggested that the Anker UPS or some other factor could be contributing to failing tests at this present time. In the 10 m chamber (which more accurately reflected testing with SLI), switching to the APC UPS and to the newer Hardware Test Utility v2 still had failing emissions at 66 MHz, and so further investigation was needed. Plans were made to analyze VxScan in the 10 m chamber without a UPS, selectively activating and deactivating key functions to try to isolate root causes of the spikes.
8	4/29/25 - 5/28/25	Standard FCC emissions testing was performed on VxScan in different configurations, without a UPS, with Hardware Test Utility v2 installed, in a 10 m chamber, under various conditions to attempt to isolate emissions related to particular functions and components. Individual features and components were disabled and disconnected to narrow down sources of emissions. Following data identifying connecting the scanner as a key source of emissions from 40-80 MHz, mitigations were tested associated with the scanner, including: • Shielding along inner walls • Ferrites applied to scanner power cable • Shielding applied to exposed surface above scanner	Al Walters, Mike Engelmann, NOVO Engineering, and Nemko San Diego; Pius Wong, Jesse DeWald, VotingWorks	The combinations of spectra obtained suggested the scanner was a source of 66 MHz spikes and other potentially concerning emissions <80 Mhz. Spectral patterns in this range suggested the amplification of harmonics of a 6 MHz signal. Ferrites/chokes were analyzed as possible mitigations. The manufacturers of the scanner and display were consulted about potential emissions sources. The scanner manufacturer confirmed there was a 6 MHz clock associated with the contact-image sensors. This could explain the spikes seen at 6 MHz apart from 48-80 MHz. The combinations of spectra obtained suggested: • Foil shielding around large areas surrounding the scanner could mitigate some of the spikes in question, but not all. • UPS usage introduced some broadband emissions but none that posed problematic QPs. • Scanner grounding configuration changes did not solve the emissions issues, and it likely was not the source.

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		- Decoupling frame and board ground on the scanner PCB - Tie shielding of power cable to frame ground - Tie metal scanner tab to frame ground		Additional analysis of the electrical system was conducted. A key finding was that pulling the VxScan scanner components outside the enclosure while leaving them electrically connected and functional resulted in passing emissions scans. This suggested that a part of the enclosure was coupling with the scanner emissions to radiate excess RF emissions, or possibly the cabling routing configuration. More investigation ruled out a variety of cable routing and ground connection features as the cause of emissions. The original cable routing and grounding features would be maintained.
9	6/9/25 - 6/10/25	NOVO Engineering investigated VxScan emissions in a 1m chamber at EMI engineering lab RF2B in San Diego, to more quickly isolate emissions sources related to the frame and develop mitigations.	Al Walters, Mike Engelmann, NOVO Engineering, and RF2B	Similar emissions spikes could be reproduced from 40-80 MHz in the 1m chamber as in the 10 m chamber, in the same VxScan configurations previously tested. The 54 MHz spur was observed using a UHF antenna to “bounce” or change when opening and closing the metal scanner access door above the scanner. Leaving it completely open mitigated some emissions, but closing it worsened emissions. Deeper investigations found that sufficient electrical contact was not made between the scanner access panel and the conductive bracket underneath it on the left side, potentially leading to poorer grounding when closed or exposing slots for RF to emit and become amplified. Pushing down on the scanner access panel could improve contact and lead to lower emissions spikes around 54 MHz.
10	6/13/25	Standard FCC emissions testing was performed on VxScan components with various configurations, including: - Improving electrical contact between scanner access panel and its supporting bracket on the left side, using copper tape and/or a weight - Application of ferrites to cabling	Al Walters, Mike Engelmann, NOVO Engineering, and Nemko San Diego; Jessed DeWald, Pius Wong, VotingWorks	The copper tape led to near elimination of the harmonic spikes of 6 MHz previously seen across the 40-80 Mhz range. This required the scanner access panel to be closed, as well as the pollworker door closed and properly sealed. After eliminating the 6 MHz harmonic spikes, a broadband “hump” was still seen in the lower frequency range, leading to a single possibly problematic QP at 60 MHz, even with the copper tape. Applying different ferrite chokes along scanner components did not mitigate this, and so further scans in different directions suggested the source of this specific noise to be at the AC power cord. Ferrite chokes offering more than 200 ohms @ 60 Mhz were applied to the AC power cord externally near the VxScan power

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				entry port, which reduced the peak of the broadband 60 MHz noise by 10 dBuV/m, leading to a passing scan. Powering with the UPS showed similar passing results. Plans were made to test again using more manufacturing-ready components and procedures, including using more robust conductive EMI gasket tape instead of copper tape, and applying a ferrite internally on the AC power lines instead of externally. Also because testing was done on a unit that already had many modifications thus far to grounding, shielding, PCB components, and parts taken in and out, a new unit was sent to NOVO Engineering for further investigation and verification of findings.
11	6/23/25	Standard FCC emissions testing was performed on a new VxScan test unit that was not previously modified for attempting to pass RF emissions testing. It would only be modified to: • Apply EMI gasket to the left underside of the scanner access panel. • Add ferrite to the internal AC power lines near the AC entry port.	Al Walters, Mike Engelmann, NOVO Engineering, and Nemko San Diego	The original emissions of interest between 40-80 MHz were no longer an issue, confirming that the root causes were primarily from lack of bridging the scanner access panel properly when closed and, secondarily, some AC power line noise. However, a new previously unobserved spike was now seen at around 30 MHz. It was noted that the APC UPS was not plugged in and fully charged at the start of the chamber testing like it is intended to be, and so it was charging while powering VxScan itself. This data was to be retaken with a properly configured APC UPS that was fully charged and plugged in before use.
12	7/10/25	Standard FCC emissions testing was performed on the VxScan unit just tested at Nemko, except that the APC UPS would now be configured properly, plugged in and fully charged for testing.	Al Walters, Mike Engelmann, NOVO Engineering, and Nemko San Diego	NOVO and Nemko reported passing results for VxScan in this configuration, both without the UPS and with the UPS (APC model), but notably without any controllers or audio devices connected to the USB ports: 

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				The 54 and 66 MHz spikes seen at SLI were not present, nor other 6 MHz harmonics. This further confirmed the root cause of failures at SLI being from improper contact on the left side of the scanner access panel, coupled with noise from the AC power input.
13	7/21/25	Standard FCC emissions certification testing was performed on a VxScan unit at Element in Longmont, CO, with SLI. An accessible controller with an audio port was included in the system, according to newly finalized test requirements.	Jessica Myers, Jesse DeWald, VotingWorks, with SLI Compliance and Element	Recommended mitigations were applied to a VxScan unit (SC-11-007) that SLI possessed since past testing in March. The Storm 9-button AudioNav controller was attached to the VxScan unit through the USB port during testing, to satisfy the latest newly settled test requirements. Prescan data showed anomalies in the high frequency range with a 5GHz spike, although this was not verified with a QP scan. Other prescan data in the low frequency range showed a potential failing spike at 720MHz. Several mitigations failed to correct the issue, including additional ferrites, changing the accessible controller and audio output setup, and grounding changes. Due to this preliminary data, and due to the new testing with the accessible controller, further emissions certification testing was cut short and postponed. Plans were made to work with another outside EMC partner to analyze the system while finalizing how to integrate accessibility features into the system.
14	8/1/25 - 11/21/25	An engineering partner (EMI Test Lab) tested emissions using four VxScan units and investigated mitigations, while VotingWorks revised the peripheral components used with the system. Additional UPS testing also indicated a change in UPS was needed.	Dennis King, Uriah Higgins, EMI Test Lab; Jesse DeWald, Pius Wong, Jessica Myers, VotingWorks	VotingWorks updated the UPS used with VxScan to the Goldenmate 1000VA/600W model. This was due to separate testing indicating that the previous UPS model (APC) would not have enough backup power in some cases, and the original UPS model (Anker) failing other EMC tests as described above. Through a series of tests, EMI Test Lab recommended mitigations for various sources of emissions in the VxScan system, including the latest spike of interest found at Element at 720 Mhz. These recommendations assumed that headphones and a PAT device were not to be provided as part of the system. Recommended mitigations to the system to address emissions were:

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				• EMI gasket tape applied to access panels, to address spikes between 54-72 MHz • Grounding wire changes, for various noise sources • USB panel mount changes, to address various intermittent noise spikes • Replacing the display with a COTS product from the same vendor that had an enclosure around the electronics, to address noise >500MHz including the 720MHz spike. • Ferrite applied to AC power line, to address noise around 195Mhz. • Increased connectivity of grounding across internal boards, to address various noise sources Near the last 11/21 test, EAC requirements for testing were formally updated so that VxScan had to include both audio output and PAT input. VotingWorks defined a plan to include headphones and a PAT device with the system, while planning to work further on mitigations that included these peripherals. The accessible audio and PAT features were planned to be separate from the USB access panel, in order to reduce emissions. VotingWorks hardware teams planned to implement the recommended changes.
15	12/19/25	VotingWorks worked with consultants to investigate emissions of a modified VxScan unit with versions of the latest recommended mitigations, and the latest requirements to include both headphones and PAT device as part of the provided system.	Jesse DeWald, VotingWorks	A modified unit was analyzed, including headphones and PAT device attached through dedicated audio and PAT ports. Ideas for mitigations were: • Increasing filtering across UPS power line and power entry module via ferrite; possibly an integrated power entry module with ferrite • SBC adjustments, including implementing a fuller enclosure and filtering of signal from the audio jack; allowing more air exchange by the SBC • Applying ferrites through all cables passing through enclosures, including for the SBC, display, and power supplies • Power supply adjustments, including possibly applying capacitors across the power supply outputs, a fuller enclosure, and ferrites; possibly

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				replacing the dual power supplies into one integrated unit.
16	2/5/26	Engineering partner EMI Test Lab tested emissions of an updated "SC-11" VxScan unit with previously recommended mitigations and accessibility peripherals (headphones, PAT device).	Dennis King, Uriah Higgins, EMI Test Lab; Jesse DeWald, Pius Wong, Jessica Myers, VotingWorks	The most updated unit marginally passed emissions in the 3m test chamber after the electrical connectivity between the two metal display support brackets was increased, bridging the connection with conductive tape. Near-field probing suggested some marginally passing QP spikes were found to have sources from the display (486, 918, and 972 MHz) and would likely decrease with even better connectivity around the display. Another marginal QP spike at 66 Mhz was inconsistently present during scans, and its source appeared to be related to the splitter & PAT cables, which could be further mitigated through a ferrite and cable configuration. One marginal QP spike at 264 MHz appeared to have its highest emissions on the right side support bracket near a PCB under a plastic hinge.
17	3/2/26 - 3/12/26	VotingWorks continued analyzing the VxScan emissions while adjusting ferrites, cable routing, and other grounding and connectivity changes, while working with a test vendor closer to VotingWorks offices. This was complementary to overall design modifications from the SC-11 version of VxScan to the newer SC-15 version, which incorporated PAT and audio interfaces more directly with the SBC. The SC-15 design also already included the upgraded COTS display from the same vendor with improved emissions and more robust electrical ground connections.	Jesse DeWald, VotingWorks, testing at CKC Laboratories, Bothell, WA	The SC-15 unit already had upgraded COTS components shown to improve emissions, but additional mitigations were still found to be needed to account for the accessibility features. Although it had better emissions than previous tests, some problematic emissions still were observed between 100-250 MHz. Appropriate mitigations to these emissions were determined to be a combination of ferrites on cables, adjusted cable routing, and additional grounding. Both external testing and internal testing corroborated passing emissions with the final set of mitigations on 5 units.
18	4/7/26, 3:39pm (Mountain Time)	Radiated emissions testing passes for unit SC-15-003 with the relevant mitigations carried over from SC-11 units.	SLI Compliance, with Jessica Myers, VotingWorks	An updated VxScan unit was retested that included mitigations for radiated RF emissions. It passed testing.

Investigative Team and Method

This section shall describe how the investigative team is assembled by the voting system manufacturer, who it consists of, and how it gathers the data to be used in the analysis. Include the RCA method employed by the manufacturer in conducting the analysis and why this method was used.

Names and Positions of members of the investigation team:
Jessica Myers, Head of Compliance
Don Chu, Lead Design Engineer
Jesse DeWald, Head of Hardware
Al Walters, Engineer; Jeff Johnson, Engineer; Mike Engelmann, Engineer (NOVO Engineering)
Pius Wong, Quality Assurance Lead
Chris Pedersen, Operations & Prototyping Support Technician
Dennis King, Engineer; Uriah Higgins, Engineer (EMC Test Lab)
Describe the data gathering process: Jessica Myers headed the investigation at first starting with the radiated emissions tests on 3/7/25, as she initiated testing at Element with SLI Compliance. Don Chu took over on 3/10/25, as he applied his expertise and experience in the design of VxScan to these tests and mitigations. Jessica and Don directly received the data from SLI, handled the devices under test when needed, and coordinated the response with other VotingWorks staff offsite. Following the initial emissions failures, Don planned for systematic investigation to find potential mitigations, isolating if they were caused more by the UPS or by VxScan components. He analyzed emissions around specific components using the near-field probe, applied shielding and other emissions-attenuating features, and swapped out components as needed to help narrow down possible sources of the failures. Chis Pedersen also was on-site at SLI and inspected test units, ruling out any production issues. Following the failed EMC tests described above (as well as several passing EMC tests), the outside engineering agency NOVO Engineering was recruited to perform more testing on the VxScan system, including using the Anker and APC UPS devices. NOVO previously tested a VxScan prototype without a UPS in July 2024, where no concerns were found in standard FCC emissions tests in a 3 m chamber using production software. This time they would also use

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the same test software utility used at SLI and conduct the failing tests from SLI in a 10 m chamber. Pius Wong coordinated getting results from NOVO to VotingWorks, also working with Jesse DeWald to provide NOVO with test equipment. Initial work used near-field probes to find potential emissions sources that could explain what was seen at Element. It suggested some inconsistency in the Anker UPS product emissions between different units, or differences in analysis methods. Analysis continued with a 10 m emissions chamber at test vendor Nemko San Diego in April through July 2025, as well as 1 m chamber at RF2B San Diego. These analyses identified problematic frequency emissions and narrowed down their sources to specific components. From this root cause data, they suggested mitigations.

Following the second tests that were cut short at Element in July 2025, the EMC engineering partner EMI Test Lab was consulted about the VxScan emissions. They worked from August 2025 to February 2026 with VotingWorks staff to analyze the system further with a fully anechoic chamber onsite. They also investigated different sets of peripherals that VotingWorks could use for integrating audio and PAT devices, as that was newly introduced partway through their testing.

After this, VotingWorks continued to investigate options for further mitigations, conducting emissions testing internally and consulting outside experts. External testing was done with CKC Laboratories in Bothell, WA, local to VotingWorks labs, in order to explore and refine the final mitigations, especially focusing on ease of production as well as final performance. Tests with CKC helped confirm internal tests about mitigations.

Describe which methodology(s) is used to conduct the root cause analysis:

The investigative team began with the “5 Why’s” strategy to discuss the causes of failures in radiated emissions tests, and to define the next courses of action in the investigations. This started by asking why the VxScan system was showing excess emissions, leading to questioning paths related to the UPS and various parts of the VxScan system.

The UPS was an early focus of the investigation due to it not being a part of past emissions tests, with a line of questioning as follows:

- Why would the Anker UPS contribute to excess radiated emissions, despite its reported FCC certification?
- Why would the Anker UPS emit more at lower frequencies <100 MHz?
- Why would the upper portion of the Anker UPS show higher levels of noise using the near-field probe?
- Why would the IoT, power control, and display electronics of the Anker UPS radiate excess emissions?
- Why would a similar Anker UPS not show the same emissions in another test setup using the near-field probe?

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These questions led to a final answer of differing build quality for the commercial off-the-shelf Anker UPS power electronics unit-to-unit. This was also confirmed by other testing on EFT resistance. As a result, the Anker UPS was deemed too unpredictable or problematic for radiated emissions and would be replaced with another UPS showing better performance. Even if this did not solve all emissions problems, it would reduce some of the sources of noise in the lower frequency range.

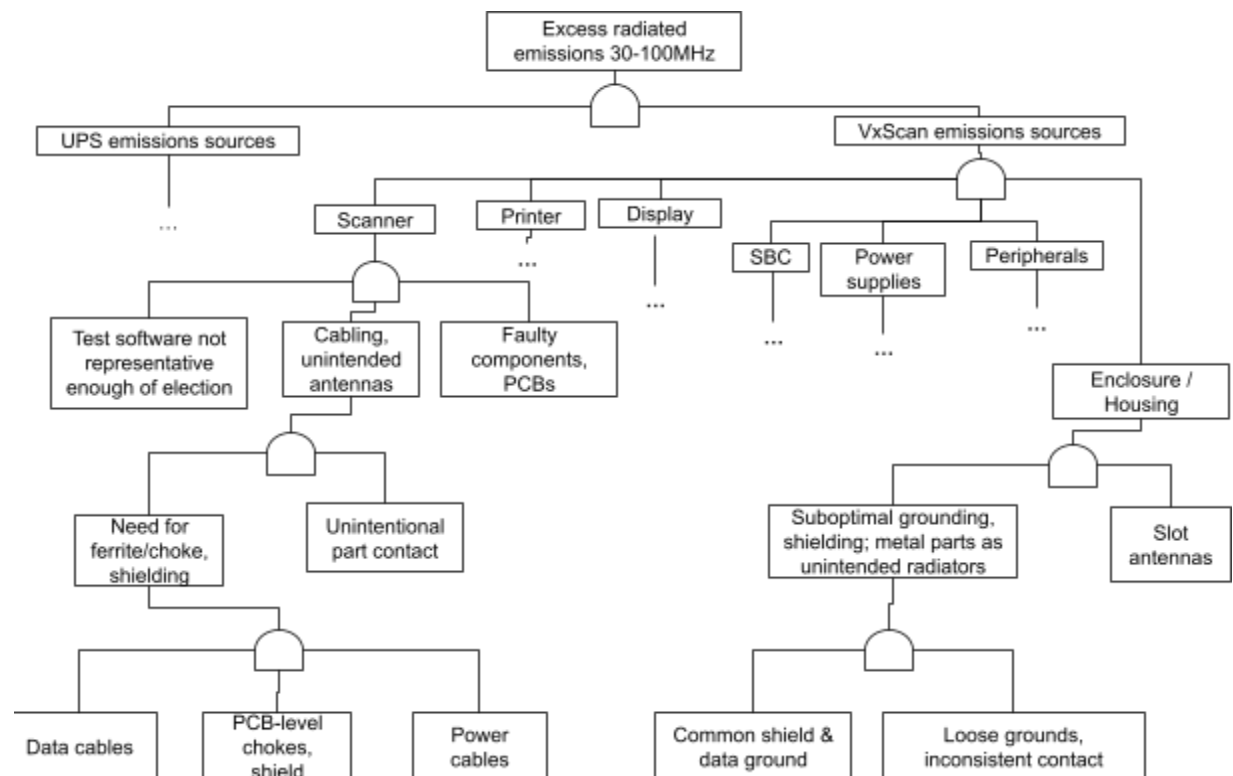
Then aspects of VxScan were investigated with similar lines of questioning, focusing on both VxScan components and on specific frequencies of interest. For example, the answers to each of these two questions led to similar intersecting investigative paths:

- Why would the printer, scanner, power components, and housing components contribute to radiated emissions at lower frequencies?

And:

- Why would VxScan emit noise <100 MHz as both resonant spikes around 48, 54, 60, 66, and 72 MHz, and also broadband noise from 40-100 MHz centered on 60 MHz?

These questions were parts of the analysis that followed the fault tree analysis pattern below. Each branch was investigated either on-site at Element, with a testing partner, or internally with VotingWorks staff, to identify the root cause of the failure seen at SLI/Element.



Ultimately the scanner, power supply, and enclosure/housing branches led to finding root causes of failures seen by SLI. However, other VxScan components were investigated, too, in

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similar detail, including the printer, display, SBC, and UPS. Components were systematically adjusted while analyzing with near-field probes and with far-field antennas in 1m, 3m, and 10m chambers. Different tools were used due to limitations in scheduling and availability, and the fault tree had to be analyzed across several months. Mitigations were developed to target the root causes in the scanner, power supply, and enclosure/housing. In the middle of this investigation, a new UPS had to be incorporated, and accessibility peripherals were added to the system, so these were also included in the fault tree analysis.

Findings and Root Cause

The primary root cause of the observed emissions failure was insufficient grounding connectivity and cable filtering around key internal electrical components. This primarily includes a small gap that sometimes appeared between the access panel doors and the support brackets underneath them, associated with emissions from 48 to 80 Mhz, 6 Mhz apart, largely seen at 54 Mhz and 66 MHz. This explains the original 54 and 66 MHz spikes seen at Element on 3/7/25. These are harmonic frequencies of 6 MHz, which is a frequency emitted by clocks associated with the contact image sensor on the scanner PCB underneath the panel. The gap at the panel reduced grounding connectivity in the top panel and may have acted as a slot antenna for these emissions.

Secondary sources of emissions that contributed to higher noise were seen at grounding wires, cable mounting connections, the display cable, and at the power input. These emissions issues were reduced when improving cable connections, adding ferrite beads, and slightly rerouting cables more consistently between units.

Tertiary root causes of the observed emissions failure were also unpredictable broadband noise contributed from peripherals. In the initial anomaly, the original Anker UPS was the source of this broadband noise. This brought emissions close to the threshold of passing/failing.

Finally, later in July 2025, the accessibility peripherals connected through USB were clear sources of failing emissions. These were newly added to the initial VxScan design, and alternate COTS accessibility peripherals were found with better emissions, when avoiding connecting through USB.

Other key findings were as follows:

1. The Anker UPS was too unpredictable or problematic for radiated emissions, as well as for other electromagnetic compatibility tests. To eliminate one likely source of electrical and emissions faults and complications, the Anker UPS was to be replaced with the Goldenmate 1000VA/600W UPS, which was shown by engineering partners to pass EMC tests.
2. The scanner PCB had clock components that were the source of 6 MHz signals that likely resulted in harmonic resonant emissions from 48 to 80 MHz seen at SLI/Element and at Nemko in far-field testing. These harmonics became coupled to scanner power and data cables, as well as the scanner access panel when it was in the closed position but not fully grounded.
3. Design changes from SC-11 to SC-15 versions incorporated upgraded COTS parts that also reduced emissions. This included a more robust display and grounding braids,

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which all were shown to reduce emissions in separate tests. This is helpful even though they were not directly causing the specific failures in the original tests at Element in March 2025. The mitigations help indirectly by lowering noise overall.

4. Past emissions tests done with NOVO Engineering/Nemko in July 2024 did not reveal the failures seen at SLI/Element in March 2025 due to at least four differences in test conditions:
 - *No UPS previously.* When the Anker UPS was present, the power electronics near the UPS display contributed to excess emissions.
 - *3 m test chamber previously.* In the 10 m chamber, emissions dropoffs did not occur as might have been extrapolated from the 3 m test data, due to more complex far-field emissions interactions and resonance from limited shielding/grounding of cables and other components.
 - *Different test software previously.* At SLI a basic Hardware Test Utility (version 1) was used that stressed all VxScan functions continuously beyond what would be seen in any election. By contrast, previous July 2024 emissions testing with NOVO/Nemko used only the production app in shoeshine mode, which still stressed the machine but more closely represented realistic continuous operation during an election. The newer Hardware Test Utility better reflects the stress of continuous operation in an election than the old version and was used for later certification tests.
 - *Different tested unit.* The actual physical unit tested in July 2024 had been built with confirmed electrical connection between the scanner access panel and the bracket underneath. Assembly variation led to more inconsistent connection in the other tested units without any EMI gasket tape.

Corrective Action(s)

The following actions were taken to address the original emissions failure, and emissions associated with the newly added PAT peripherals:

1. Apply EMI gasket tape under the left side of the scanner access panel to increase connection with the support bracket underneath. This can be compressive conductive gasket material meant for repeated industrial use, such as the tested component Wurth 3021005, with an appropriate height between 0.010" to 0.030". This reduces the major low frequency emissions spikes seen originally. See the images below for example applications:



2. Replace the USB bulkhead cables with a more robust version that has better 360° connectivity with its mounting plate (L-com ECF504-30-1AAS). This reduces intermittent noise related to the original USB cable not contacting chassis ground around the entire bulkhead. The additional ground connectivity of the new cable reduces emissions across a lower range seen in the original anomaly and other testing.
3. Apply ferrite beads on the USB-C cable to the display, the AC power input line, the headphone cable near the jack, and the PAT cable near the jack. The ferrites have rated impedances of 400-600Ohm @ 100MHz to account for the frequencies seen radiating from them in additional testing.
4. Replace the backup UPS used with VxScan with the Goldenmate 1000VA/600W model. This had more predictable emissions than the Anker while also supporting other needs unrelated to emissions related to backup power and user experience.

These mitigations are reflected in the latest SC-15 VxScan hardware submitted for certification testing. This SC-15 hardware addresses the concerns seen in the original SC-11 build.

Solution Management

The purpose of this section is to manage the corrective action(s) moving forward. This should detail all process changes to manage those corrective actions, and steps taken to ensure the actions eliminate the anomaly over time.

Plans to manage the solution are as follows:

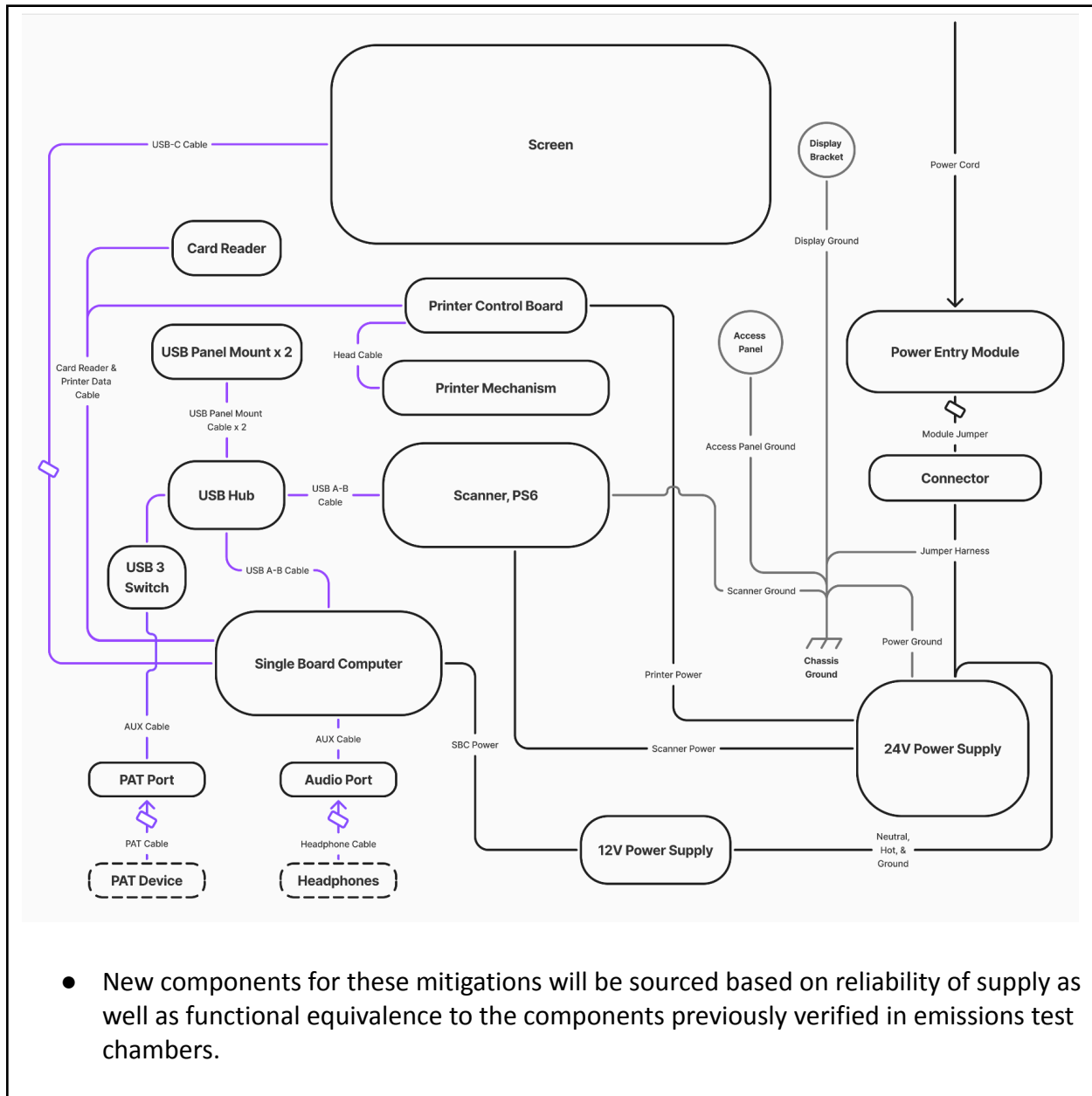
Testing and verification:

- Performance of modified designs was verified through both internal and external testing before retesting with SLI/Element.
- Testing software for continuous operation during electrical testing was updated for more realistic simulation of elections. It no longer prints an unreasonable amount of paper every few seconds, and it allows more control of individual VxScan functions. This allows for more useful test data.

Design & Production:

- Design and assembly plans will be modified to incorporate the addition of the EMI gasket, new bulkhead cables, and ferrites. These were incorporated into the latest SC-15 version of the VxScan design, which already incorporated accessibility peripherals and an upgraded display. Placement of ferrites is also indicated in the updated electrical cabling diagram of the system shown below:

EAC (Election Assistance Commission) Root Cause Analysis



- New components for these mitigations will be sourced based on reliability of supply as well as functional equivalence to the components previously verified in emissions test chambers.